

Phase 6 - 9 Documentation (Enrollment & Attendance Tracker)

Phase 6: User Interface Development

Lightning App Builder -> Record Pages

Setup → Lightning App Builder → New → Record Page → Object: Attendance__c → Add Highlights Panel, Related Lists (Student, Course, Status, Date).

The screenshot shows the Salesforce Lightning App Builder interface. The top navigation bar includes 'Lightning App Builder', 'Pages', and 'Attendance Record Page'. The main workspace displays a preview of the 'Attendance Record Page' for a record named 'Shruti Misal'. The left sidebar shows a list of components under the 'Standard (41)' category, including 'Highlights Panel', 'Related Lists', and 'Table'. The right sidebar shows the 'Page' configuration panel with fields for 'Label', 'API Name', 'Page Type', 'Object', 'Template', and 'Description'. The 'Object' field is set to 'Attendance'.

Date	Status	Enrollment	Marked By
9/4/2025	Present	ENR-2023	Harshad Anasure

Tabs (Custom Object Tabs)

Setup → Tabs → New → Custom Object Tab → Attendance Records

Setup → Tabs → New → Custom Object Tab → Courses

Setup → Tabs → New → Custom Object Tab → Enrollments

Setup → Tabs → New → Custom Object Tab → Grades

Setup → Tabs → New → Custom Object Tab → Guardians

Setup → Tabs → New → Custom Object Tab → Integration Logs

Setup → Tabs → New → Custom Object Tab → Students

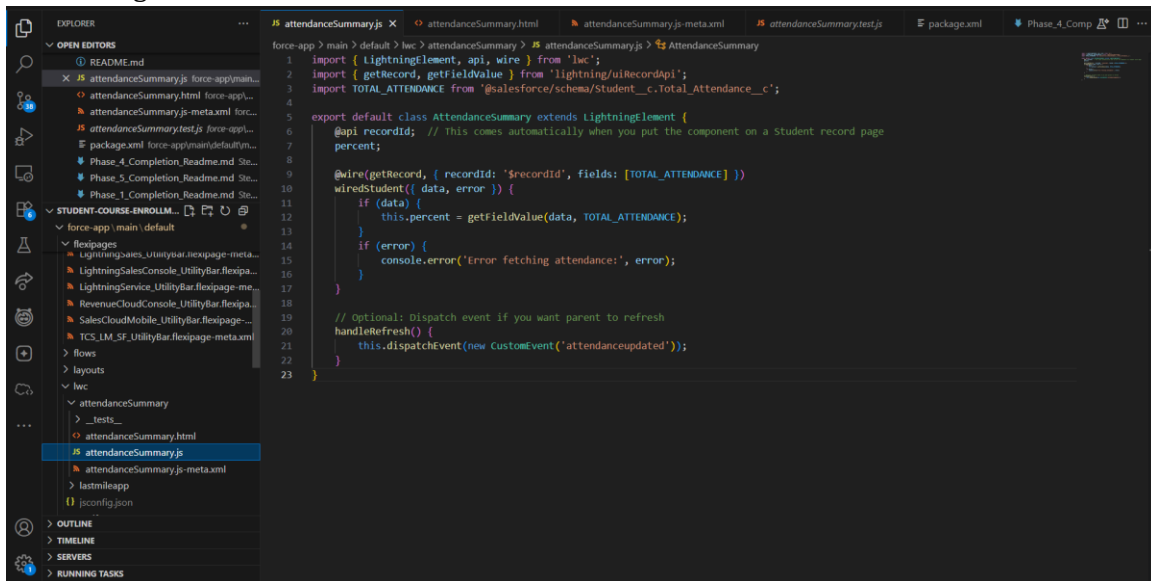
The screenshot shows the 'Attendance Records' custom object tab configuration page in Salesforce Setup. The left sidebar has a search bar with 'tabs' and a 'User Interface' section with 'Rename Tabs and Labels' and 'Tabs'. The main content area is titled 'Edit Custom Object Tab Attendance Records' and includes a 'Help for this Page' link. Below the title is a section 'Custom Tab Definition Edit' with a 'Required Information' indicator. It contains fields for 'Tab Label' (Attendance Records), 'Object' (Attendance), and 'Tab Style' (Books). There is also an optional field for 'Splash Page Custom Link' set to '-None-' and a 'Description' text area. At the bottom are 'Save' and 'Cancel' buttons.

The screenshot shows the 'Custom Tabs' configuration page in Salesforce Setup. The left sidebar is identical to the previous screenshot. The main content area is titled 'Custom Tabs' and includes a 'Help for this Page' link. Below the title is a section 'Custom Object Tabs' with a 'New' button and a 'What Is This?' link. It contains a table with columns 'Action', 'Label', 'Tab Style', and 'Description'. The table lists several custom object tabs: Attendance Records (Books), Courses (Bank), Enrollments (Airplane), Grades (Leaf), Guardians (Red Cross), Integration Logs (Stipewitch), and Students (Computer). Below the table is a section 'Web Tabs' with a 'New' button and a 'What Is This?' link, and a message 'No Web Tabs have been defined'.

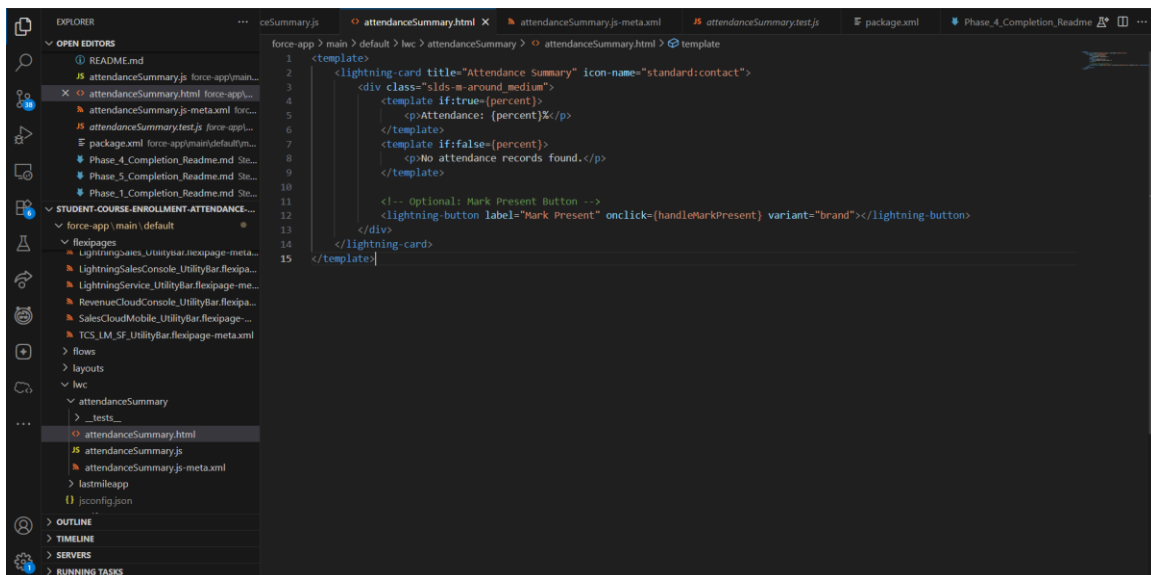
Action	Label	Tab Style	Description
Edit Del	Attendance Records	Books	
Edit Del	Courses	Bank	
Edit Del	Enrollments	Airplane	
Edit Del	Grades	Leaf	
Edit Del	Guardians	Red Cross	
Edit Del	Integration Logs	Stipewitch	
Edit Del	Students	Computer	

LWC (Lightning Web Components)

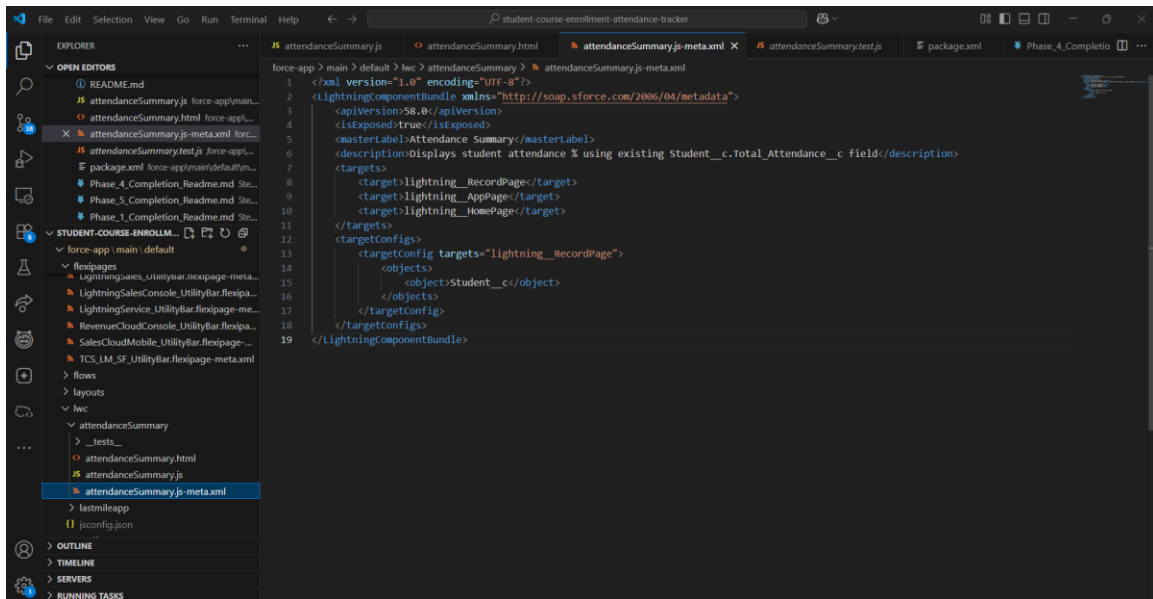
VS Code → Create LWC → attendanceSummary → Deploy to Salesforce → Add to Attendance Record Page.



```
1 import { LightningElement, api, wire } from 'lwc';
2 import { getRecord, getFieldValue } from 'lightning/uiRecordApi';
3 import TOTAL_ATTENDANCE from '@salesforce/schema/Student__c.Total_Attendance__c';
4
5 export default class AttendanceSummary extends LightningElement {
6     @api recordId; // This comes automatically when you put the component on a Student record page
7     percent;
8
9     @wire(getRecord, { recordId: '$recordId', fields: [TOTAL_ATTENDANCE] })
10     wiredStudent({ data, error }) {
11         if (data) {
12             this.percent = getFieldValue(data, TOTAL_ATTENDANCE);
13         }
14         if (error) {
15             console.error('Error fetching attendance:', error);
16         }
17     }
18
19     // Optional: Dispatch event if you want parent to refresh
20     handleRefresh() {
21         this.dispatchEvent(new CustomEvent('attendanceupdated'));
22     }
23 }
```



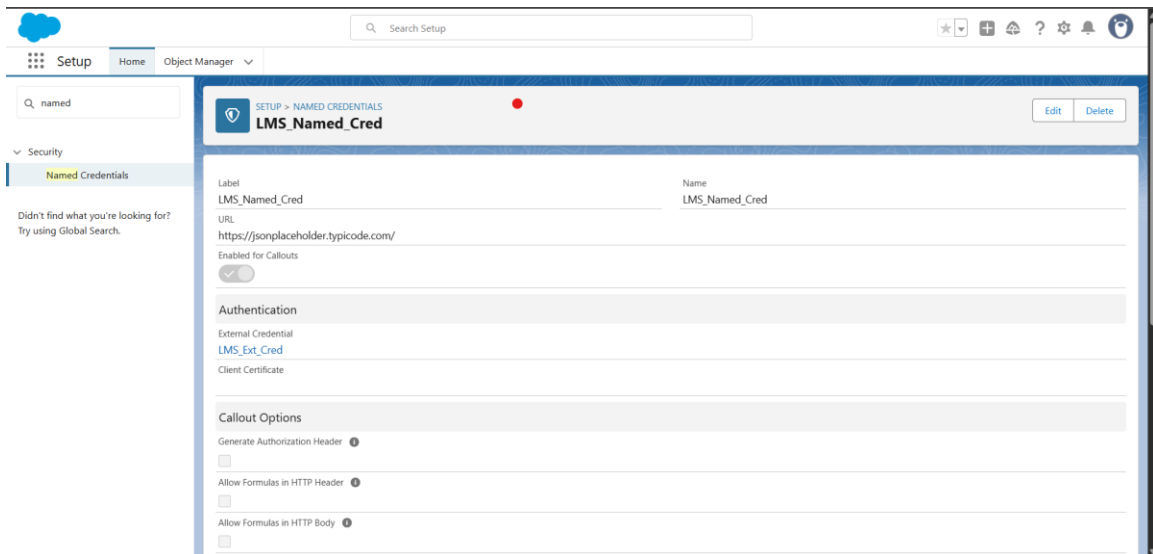
```
1 <template>
2     <lightning-card title="Attendance Summary" icon-name="standard:contact">
3         <div class="slds-w-around_medium">
4             <template if:true={percent}>
5                 <p>Attendance: {percent}%</p>
6             </template>
7             <template if:false={percent}>
8                 <p>No attendance records found.</p>
9             </template>
10
11             <!-- Optional: Mark Present Button -->
12             <lightning-button label="Mark Present" onclick={handleMarkPresent} variant="brand"></lightning-button>
13         </div>
14     </lightning-card>
15 </template>
```



Phase 7: Integration & External Access

Named Credentials

Setup → Named Credentials → New → Label: LMS_Named_Cred → URL:
<https://jsonplaceholder.typicode.com/> → External Credential: LMS_Ext_Cred → Save.



External Credentials

Setup → External Credentials → New → Label: LMS_Ext_Cred → Authentication Protocol: Basic Authentication → Save.

The screenshot shows the Salesforce Setup interface for managing external credentials. The left sidebar includes a search bar and a navigation menu with 'Security' and 'Named Credentials'. The main content area is titled 'LMS_Ext_Cred' and contains the following sections:

- Label:** LMS_Ext_Cred
- Name:** LMS_Ext_Cred
- Authentication Protocol:** Basic Authentication
- Managed Package Access:** (Empty section)
- Created By:** Namespace
- Related Named Credentials:** A table with columns Label, Name, and URL. It contains one entry: LMS_Named_Cred, LMS_Named_Cred, and https://jsonplaceholder.typicode.com/.
- Principals:** A table with columns Sequence, Parameter Name, Identity Type, Authentication Status, Username, and Actions. It contains one entry: 1, ReqRes_Principal, Named Principal, Configured, testuser, and a dropdown arrow.

The screenshot shows the Salesforce IDE with the following Apex code:

```
1 public with sharing class LmsSimpleGet {
2
3     @AuraEnabled
4     public static String getCourses() {
5         HttpRequest req = new HttpRequest();
6         req.setEndpoint('callout:LMS_Named_Cred/courses'); // relative path
7         req.setMethod('GET');
8
9         Http http = new Http();
10        HttpResponse res = http.send(req);
11
12        return res.getBody();
13    }
14 }
15 }
```

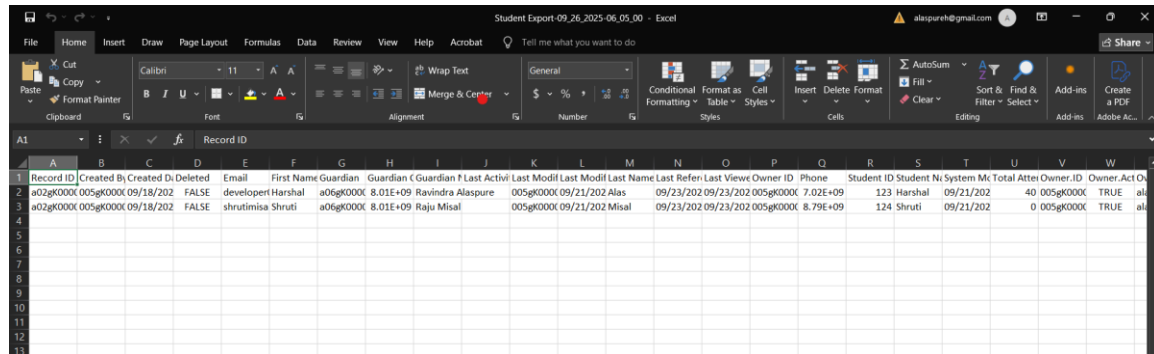
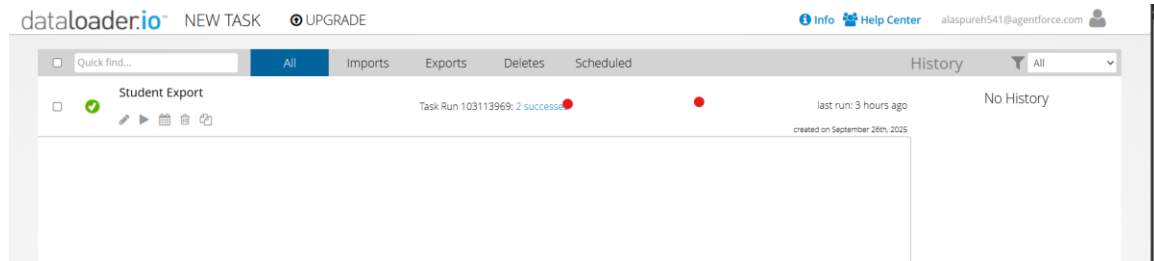
Phase 8: Data Management & Deployment

Data Loader

Open Data Loader → Login with Salesforce → Select Insert → Object: Student__c → Choose CSV file → Map fields → Finish.

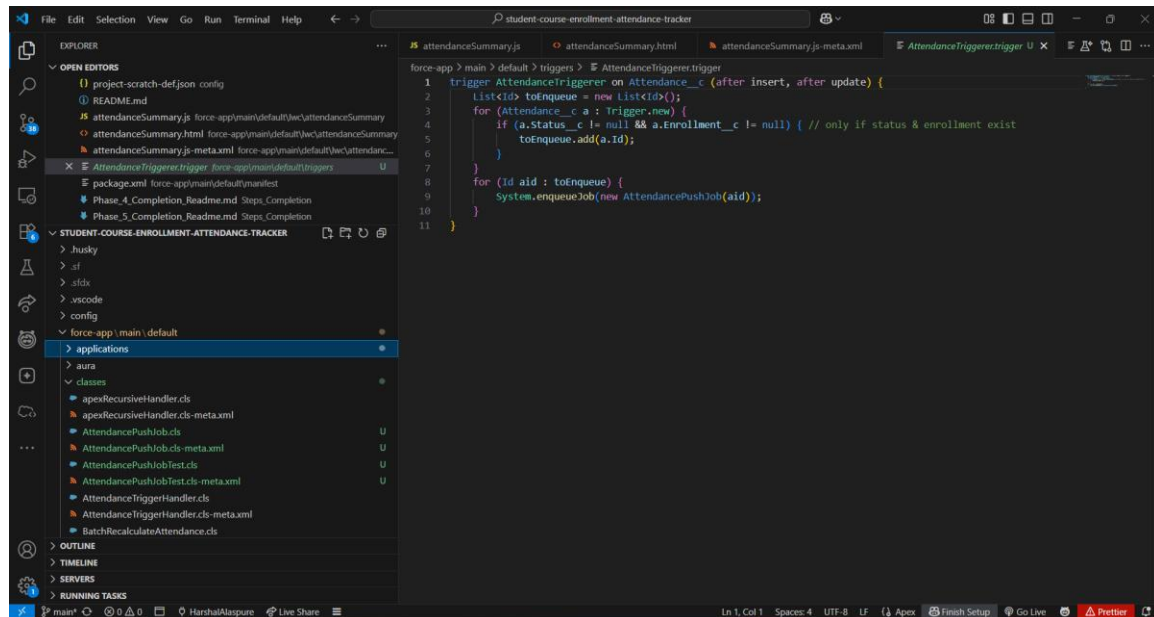
Data Export & Backup

Setup → Data Export → Schedule Export → Select Student_c → Export to CSV → Download.



VS Code & SFDX

Install VS Code → Install Salesforce Extension Pack → Connect to Org (sfdx force:auth:web:login) → Push/Deploy metadata.



Phase 9: Reporting, Dashboards & Security Review

Reports

Reports

Recent

5 items

REPORTS	Report Name	Description	Folder	Created By	Created On	Subscribed
Recent	Sample Flow Report: Screen Flows	Which flows run, what's the status of each interview, and how long do users take to complete the screens?	Public Reports	Automated Process	7/17/2025, 9:17 AM	
Created by Me	Grades Reports		Private Reports	Harshal Alaspure	9/26/2025, 12:52 AM	
Private Reports	First Attendance Record		Private Reports	Harshal Alaspure	9/26/2025, 12:37 AM	
Public Reports	Enrolled Students Report		Private Reports	Harshal Alaspure	9/26/2025, 12:49 AM	
All Reports	Attendance Records of students	Sorted by date in ascending here are the attendance reports	Private Reports	Harshal Alaspure	9/26/2025, 12:43 AM	
FOLDERS						
All Folders						
Created by Me						
Shared with Me						
FAVORITES						
All Favorites						

Reports → New Report → Object: Enrollment_c → Report: Enrolled Students Report

Report: Students with Enrollments and Courses

Enrolled Students Report

Enable Field Editing

Total Records: 3, Average Total Attendance: 66.67

Student: Student Name	Last Name	Enrollment: Enrollment Number	Course: Course Name	Email	Guardian Name	Total Attendance	Status	
1	Harshal	Alaspure	ENR-0001	Data Structures	[REDACTED]	Ravindra Alaspure	33.33	Enrolled
2	Harshal	Alaspure	ENR-0002	Machine Learning	[REDACTED]	Ravindra Alaspure	33.33	Enrolled
3	Shruti	Misal	ENR-0003	Machine Learning	[REDACTED]	Raju Misal	100.00	Enrolled
4						Avg: 66.67		

Reports → New Report → Object: Attendance_c → Report: Attendance Records of students

Report: Attendance Records with Enrollment

Attendance Records of students

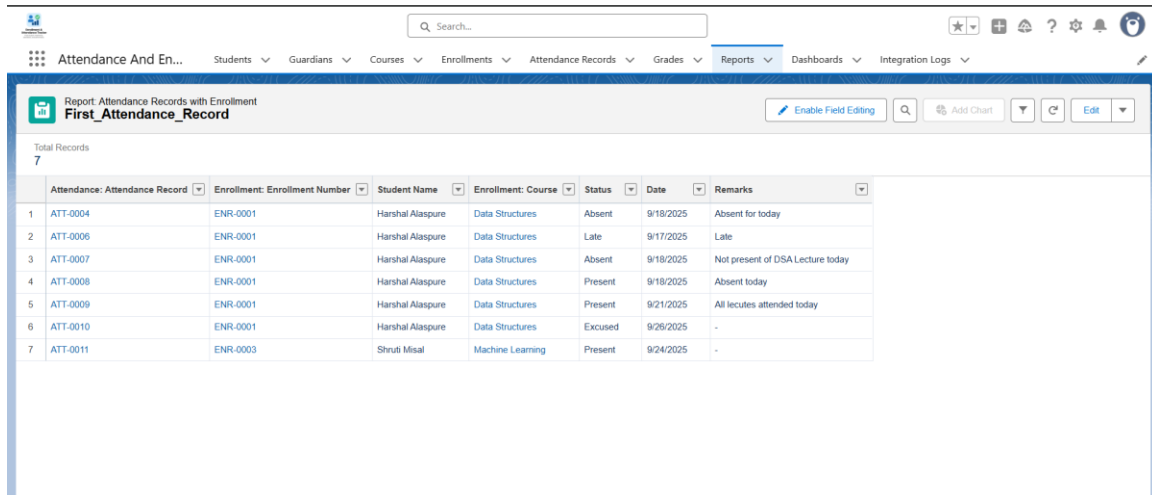
Sorted by date in ascending here are the attendance reports

Enable Field Editing

Total Records: 7

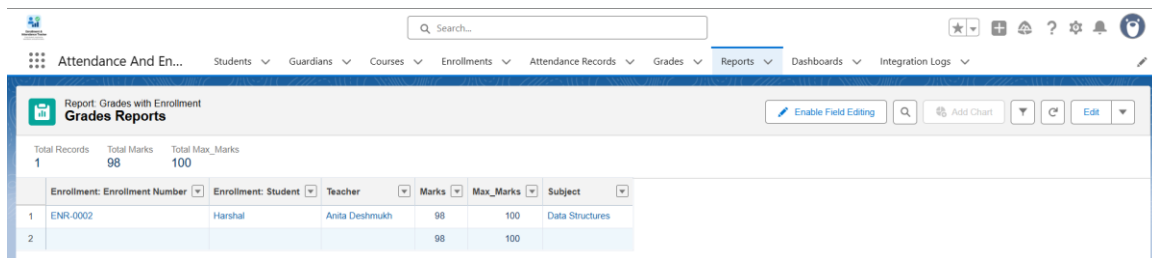
Date	Student Name	Enrollment: Course	Status	Remarks
9/17/2025	Harshal Alaspure	Data Structures	Late	Late
9/18/2025	Harshal Alaspure	Data Structures	Absent	Absent for today
9/18/2025	Harshal Alaspure	Data Structures	Absent	Not present of DSA Lecture today
9/18/2025	Harshal Alaspure	Data Structures	Present	Absent today
9/21/2025	Harshal Alaspure	Data Structures	Present	All lectures attended today
9/24/2025	Shruti Misal	Machine Learning	Present	-
9/26/2025	Harshal Alaspure	Data Structures	Excused	-

Reports → New Report → Object: Attendance_c → Report: First_Attendance_Record



	Attendance: Attendance Record	Enrollment: Enrollment Number	Student Name	Enrollment: Course	Status	Date	Remarks
1	ATT-0004	ENR-0001	Harshal Alaspure	Data Structures	Absent	9/18/2025	Absent for today
2	ATT-0006	ENR-0001	Harshal Alaspure	Data Structures	Late	9/17/2025	Late
3	ATT-0007	ENR-0001	Harshal Alaspure	Data Structures	Absent	9/18/2025	Not present of DSA Lecture today
4	ATT-0008	ENR-0001	Harshal Alaspure	Data Structures	Present	9/18/2025	Absent today
5	ATT-0009	ENR-0001	Harshal Alaspure	Data Structures	Present	9/21/2025	All lectures attended today
6	ATT-0010	ENR-0001	Harshal Alaspure	Data Structures	Excused	9/26/2025	-
7	ATT-0011	ENR-0003	Shruti Misal	Machine Learning	Present	9/24/2025	-

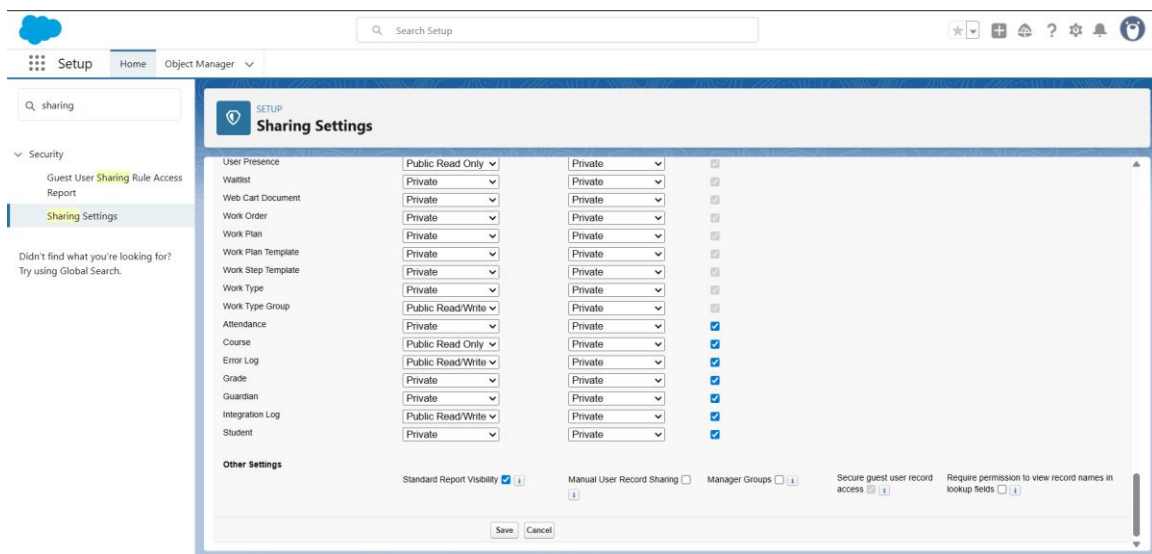
Reports → New Report → Object: Grade_c → Report: Grades Reports



	Enrollment: Enrollment Number	Enrollment: Student	Teacher	Marks	Max_Marks	Subject
1	ENR-0002	Harshal	Anita Deshmukh	98	100	Data Structures
2				98	100	

Sharing Settings (OWD)

Setup → Sharing Settings → Edit OWD → Define object-level access for Attendance, Enrollment, Grade, etc.



Object	Public Read Only	Private	Public Read/Write	Private
User Presence	☐	☐	☐	☐
Waitlist	☐	☐	☐	☐
Web Cart Document	☐	☐	☐	☐
Work Order	☐	☐	☐	☐
Work Plan	☐	☐	☐	☐
Work Plan Template	☐	☐	☐	☐
Work Step Template	☐	☐	☐	☐
Work Type	☐	☐	☐	☐
Work Type Group	☐	☐	☐	☐
Attendance	☒	☐	☐	☐
Course	☐	☐	☐	☐
Error Log	☐	☐	☐	☐
Grade	☐	☐	☐	☐
Guardian	☐	☐	☐	☐
Integration Log	☐	☐	☐	☐
Student	☐	☐	☐	☐

Other Settings

Standard Report Visibility ☒ Manual User Record Sharing ☐ Manager Groups ☐ Secure guest record access ☐ Require permission to view record names in lookup fields ☐

Field Level Security

Setup → Object Manager → [Selecting Object] → Fields & Relationships → Select Field → Set Field-Level Security.

The screenshot shows the Salesforce Setup interface. The left sidebar has a search bar with "audit" and a "Security" section with "View Setup Audit Trail". The main content area is titled "Set Field-Level Security Email". Below the title is a table with columns "Field Label", "Email", and "Data Type". The "Data Type" column shows "Email". Below the table is a section titled "Field-Level Security for Profile" with a table of profiles and their security settings.

Field-Level Security for Profile	Visible	Read Only
Analytics Cloud Integration User	☒	☐
Analytics Cloud Security User	☒	☐
Anypoint Integration	☒	☐
Contract Manager	☒	☐
Cross Org Data Proxy User	☒	☐
Custom: Marketing Profile	☒	☐
Custom: Sales Profile	☒	☐
Custom: Support Profile	☒	☐
Einstein Agent User	☒	☐
Force.com - App Subscription User	☒	☐

Session Settings

Setup → Session Settings → Session Timeout = 30 minutes → Save.

The screenshot shows the Salesforce Setup interface for Session Settings. The left sidebar has a search bar with "session" and a "Session Management" section with "Session Settings". The main content area is titled "Session Settings" and contains a "Session Timeout" section with a "Timeout Value" of "30 minutes". Below this is a "Session Settings" section with various checkboxes.

Session Timeout

Timeout Value: 30 minutes

☐ Disable session timeout warning popup

☒ Force logout on session timeout

Session Settings

☐ Lock sessions to the IP address from which they originated

☒ Lock sessions to the domain in which they were first used

☐ Terminate all of a user's sessions when an admin resets that user's password

☒ Force relogin after Login-As-User

☐ Require HttpOnly attribute

☐ Use POST requests for cross-domain sessions

☐ Enforce login IP ranges on every request

☐ When embedding a Lightning application in a third-party site, use a session token instead of a session cookie.

Login IP Ranges

Setup → Profiles → [Selecting Profile] → Login IP Ranges → No restrictions added.

Audit Trail

Setup → View Setup Audit Trail → Download CSV if required.

Setup

Home

Object Manager

Search Setup

☆

+

🏠

?

🔔

👤

Q audit

Security

View Setup Audit Trail

Didn't find what you're looking for?

Try using Global Search.

VIEW

SETUP

View Setup Audit Trail

View Setup Audit Trail

Date	User	Source Namespace Prefix	Action	Section	Delegate User
9/26/2025, 1:23:21 AM PDT	alaspureh541@agentforce.com		Changed Session Timeout Value from 120 to 30 minutes	Session Settings	
9/26/2025, 1:23:21 AM PDT	alaspureh541@agentforce.com		Session Security Level for Multi-Factor Authentication was set to High Assurance	Session Settings	
9/26/2025, 1:23:21 AM PDT	alaspureh541@agentforce.com		Session Security Level for Passwordless Login was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alaspureh541@agentforce.com		Session Security Level for Lightning Login was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alaspureh541@agentforce.com		Session Security Level for Activation was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alaspureh541@agentforce.com		Session Security Level for Delegated Authentication was set to Standard	Session Settings	
9/26/2025, 1:23:21 AM PDT	alaspureh541@agentforce.com		Session Security Level for Username Password was set to Standard	Session Settings	
9/25/2025, 10:35:59 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:33:32 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:32:21 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:29:43 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:26:23 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:20:22 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:18:42 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:16:29 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:15:17 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:11:14 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:06:26 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:03:42 PM PDT	alaspureh541@agentforce.com		Changed AttendancePushJobTest Apex Class code	Apex Class	
9/25/2025, 10:02:03 PM PDT	alaspureh541@agentforce.com		Created AttendancePushJobTest Apex Class code	Apex Class	

Download setup audit trail for last six months (Excel .csv file)